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Applied Machine Learning Project

Prediction of High School Students' GPA

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1 Introduction

The Finnish youth have become progressively weaker in mathematics in comparison to the top year in history, 2006, according to the Pisa-research of 2022 (Opetus- ja kulttuuriministeriö, 2022). Points in literacy and natural sciences have also decreased from the year 2006. Although the data set in this project is not collected from Finnish high schools, there might be usefulness to research the performance of Finnish high school students in the future the same way. Kallunki (2023) has listed some worrying development in the Finnish elementary schools, as some ninth graders have a math competence of a sixth grader. For us to become competitive in the international world, we need to be excellent performers in school.

In this project, data of 2392 high school students' performance through GPA will be analyzed and inferred using machine learning models, such as Python's Scikit-Learn library's ExtraTreesRegressor and MLPRegressor. The dataset is from Kaggle and has been put together by Rabie El Kharoua (Kharoua, 2024). The variables of the dataset include StudentID, Age, Gender, Ethnicity, Parental Education, Weekly Study Times, Absences, Tutoring, Parental Support, Extracurricular activities, Sports, Music, Volunteering, GPA and Grade Class. More information, for example about the categories of the variables, can be found from the dataset's website.

Although Salem et. al. (2013) have examined the factors of medical school students' academic performance, their results can be compared with the results of this project. Their results conclude that age, motivation, types of learning resources, study time and type of transportation used have shown to create a significant difference in GPA between males and females. Bilal et. al. (2022) have also proved, that in matriculation, students with marks greater than 69% in Biology or students with marks greater than 91% in Islamiat, were likely to fall in high performers in the final semester. In addition, in intermediate, students with marks less than or equal to 58% in English, were likely to have fall in low performers in the last semester.

Moreover, the justification of the project can be stated to be the very good accuracy and predictive capability of the former research. The research of Bilal et. al. (2022) used a support vector machine, which yielded a 92% accuracy and a random forest, which yielded an 86% accuracy. Salem et. al. (2013) used SPSS in their research, but yielded significant results, which at the very least encourages the use of statistical inference in student performance estimation. Oku and Sato (2021) have also utilized random forest quite successfully, yielding a 67% area of the ROC curve.

2 The machine learning process and model building

I start the data extraction by reading the file into the Jupyter Notebook, which contains 2392 rows and 15 columns. The idea of the project is to predict GPA with students' other features, i. e. other features will be predictors. There are two features, which won't be needed, StudentID and GradeClass indicating the students' GPA class. GradeClass would be needed for a classification model, however this time I will be regressing GPA. Here is how the feature matrix X (Figure 1) and target matrix y (Figure 2) look like:

	Age	Gender	Ethnicity	ParentalEducation	StudyTimeWeekly	Absences	Tutoring	ParentalSupport	Extracurricular	Sports	Music	Volunteering
0	17	1	0	2	19.833723	7	1	2	0	0	1	0
1	18	0	0	1	15.408756	0	0	1	0	0	0	0
2	15	0	2	3	4.210570	26	0	2	0	0	0	0
3	17	1	0	3	10.028829	14	0	3	1	0	0	0
4	17	1	0	2	4.672495	17	1	3	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...
2387	18	1	0	3	10.680555	2	0	4	1	0	0	0
2388	17	0	0	1	7.583217	4	1	4	0	1	0	0
2389	16	1	0	2	6.805500	20	0	2	0	0	0	1
2390	16	1	1	0	12.416653	17	0	2	0	1	1	0
2391	16	1	0	2	17.819907	13	0	2	0	0	0	1

2392 rows × 12 columns

Figure 1 Feature matrix X

```
0       2.929196
1       3.042915
2       0.112602
3       2.054218
4       1.288061
          ...
2387    3.455509
2388    3.279150
2389    1.142333
2390    1.803297
2391    2.140014
Name: GPA, Length: 2392, dtype: float64
```

Figure 2 Target matrix y

The feature matrix has now the same number of rows, but only 12 columns. Before splitting the data to training and testing sets, I want to highlight the correlation matrix and heatmap:

	StudentID	Age	Gender	Ethnicity	ParentalEducation	StudyTimeWeekly	Absences	Tutoring	ParentalSupport	Extracurricular	Sp
StudentID	1.000000	-0.042255	-0.014625	-0.012990	-0.002307	0.026976	0.014841	-0.007834	0.003016	-0.003611	-0.02(
Age	-0.042255	1.000000	0.044895	-0.028473	0.025099	-0.006800	-0.011511	-0.012076	0.033197	-0.025061	-0.04(
Gender	-0.014625	0.044895	1.000000	0.016010	0.006771	0.011469	0.021479	-0.031597	0.008065	-0.005964	-0.00(
Ethnicity	-0.012990	-0.028473	0.016010	1.000000	0.033595	0.007184	-0.025712	-0.017440	0.020922	-0.008927	-0.00-
ParentalEducation	-0.002307	0.025099	0.006771	0.033595	1.000000	-0.011051	0.036518	-0.017340	-0.017463	0.007479	0.00:
StudyTimeWeekly	0.026976	-0.006800	0.011469	0.007184	-0.011051	1.000000	0.009326	0.028930	0.035800	-0.022860	0.00(
Absences	0.014841	-0.011511	0.021479	-0.025712	0.036518	0.009326	1.000000	-0.015534	0.002108	0.000360	0.04'
Tutoring	-0.007834	-0.012076	-0.031597	-0.017440	-0.017340	0.028930	-0.015534	1.000000	-0.000824	0.004865	0.00(
ParentalSupport	0.003016	0.033197	0.008065	0.020922	-0.017463	0.035800	0.002108	-0.000824	1.000000	-0.008381	-0.00(
Extracurricular	-0.003611	-0.025061	-0.005964	-0.008927	0.007479	-0.022860	0.000360	0.004865	-0.008381	1.000000	-0.01'
Sports	-0.020703	-0.046320	-0.008897	-0.004484	0.002029	0.006836	0.041454	0.006278	-0.006176	-0.011820	1.00(
Music	-0.005468	-0.003492	0.007109	-0.014627	0.039439	0.007791	-0.008692	-0.011385	0.035122	-0.014191	-0.02(
Volunteering	0.008011	0.013074	-0.000200	0.013468	0.011960	-0.016604	-0.018528	-0.050898	-0.006036	-0.007427	-0.00:
GPA	-0.002697	0.000275	-0.013360	0.027760	-0.035854	0.179275	-0.919314	0.145119	0.190774	0.094078	0.05'
GradeClass	-0.098500	-0.006250	0.022998	-0.023326	0.041031	-0.134131	0.728633	-0.111695	-0.136823	-0.069733	-0.02(

Figure 3 Correlation matrix

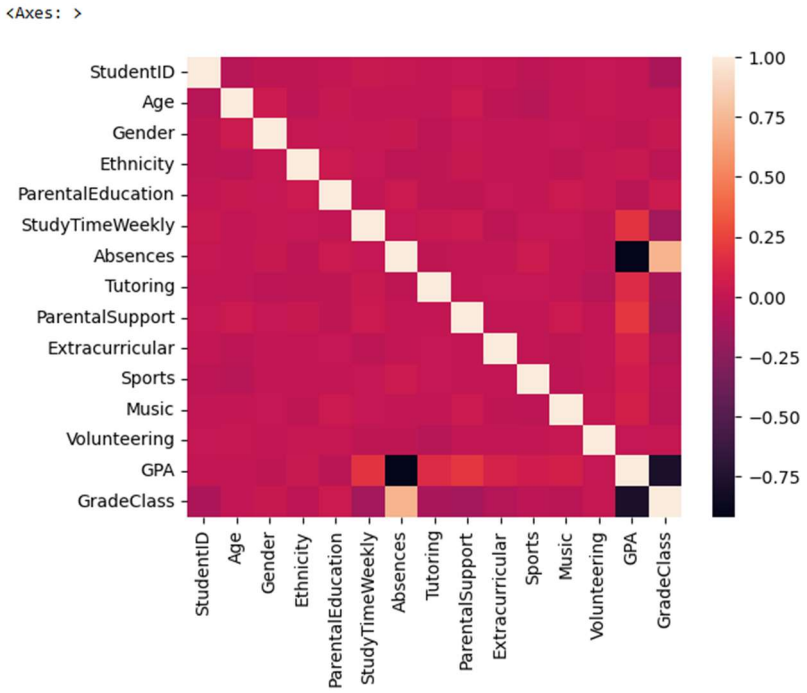


Figure 4 Correlation heatmap

The correlation between other features is closer to zero, except GPA and Absences (Figures 3 and 4). It will be apparent in the feature selection, but Absences will be one of the

most important features when selecting a subset of the most variance causing features from the model. But more on that later.

I split the data to a train and test set, 0.8 and 0.2 respectively. The algorithm that I selected for the prediction of GPA is ExtraTreesRegressor. The justification for the algorithm can, for example, be found in Irshada's and Kumar's (2023) research, where the ETR was used to a high accuracy with the Random Forest model. It is also emphasized, that there are more optimizable parameters than in a normal decision tree and that the threshold rules are selected at random for randomly selected features, after which the best thresholding rules are voted (Välisuo, 2023). It is also easy to draw the most important features from the data set with the algorithm.

The number of estimators for the ETR (ExtraTreesRegressor) is at the default 100. After I fit the training data to the model, it yields a cross-validation score of 0.93 and a 0.92 accuracy on the test set. The scores are very good, so basically it is not necessary to drop the least important features. I'll do it nonetheless for the sake of comparison. Before that though, I'll tune the parameters of the model with GridSearchCV to select the best subset for prediction. The parameter tuning yields the best ETR estimator, which has a max depth of 9 and 24 estimators. The tuned ETR has a cross-validation score of 0.93 and a 0.92 accuracy on the test set, i.e. they are quite the same with the earlier model.

Next, the most important features of the data set will be plotted and five of the most important ones selected for the model.

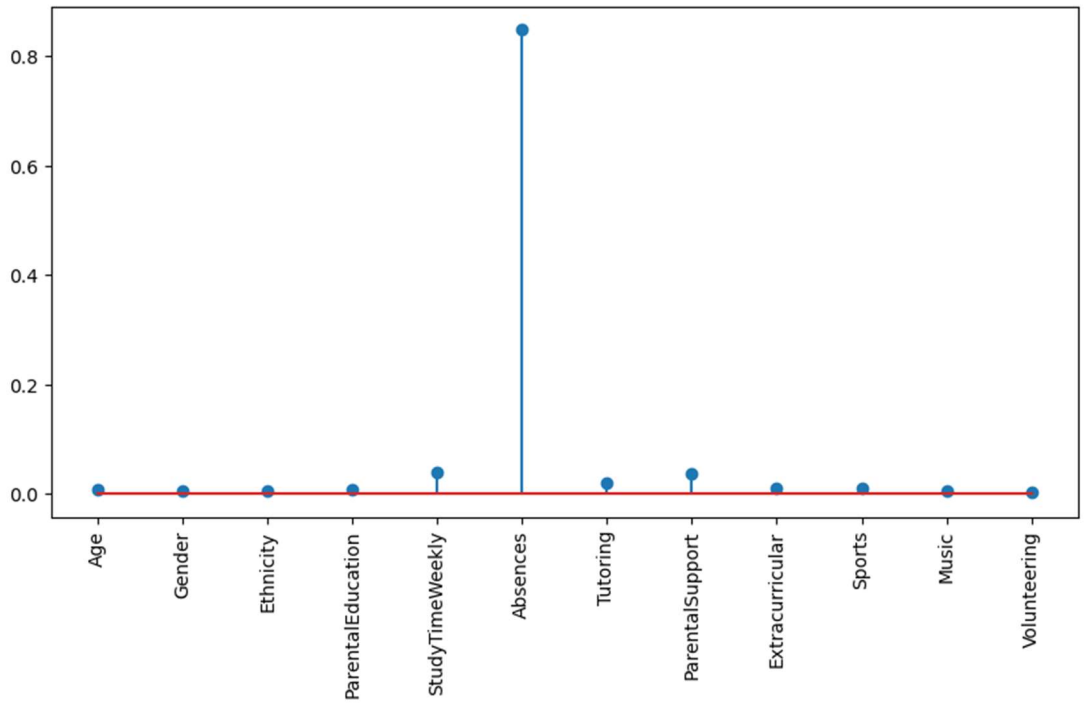


Figure 5 Feature importances

The most important features in descending order are Absences, StudyTimeWeekly, ParentalSupport, Tutoring and Extracurricular (Figure 5). Important to note, Absences was -0.92, StudyTimeWeekly 0.18, ParentalSupport 0.19, Tutoring 0.145, and Extracurricular 0.09 in the correlation matrix. Therefore, according to the correlation matrix, all features except Absence affect GPA positively. To yet visualize the relation of Absences against GPA, I plot it on a histogram plot chart:

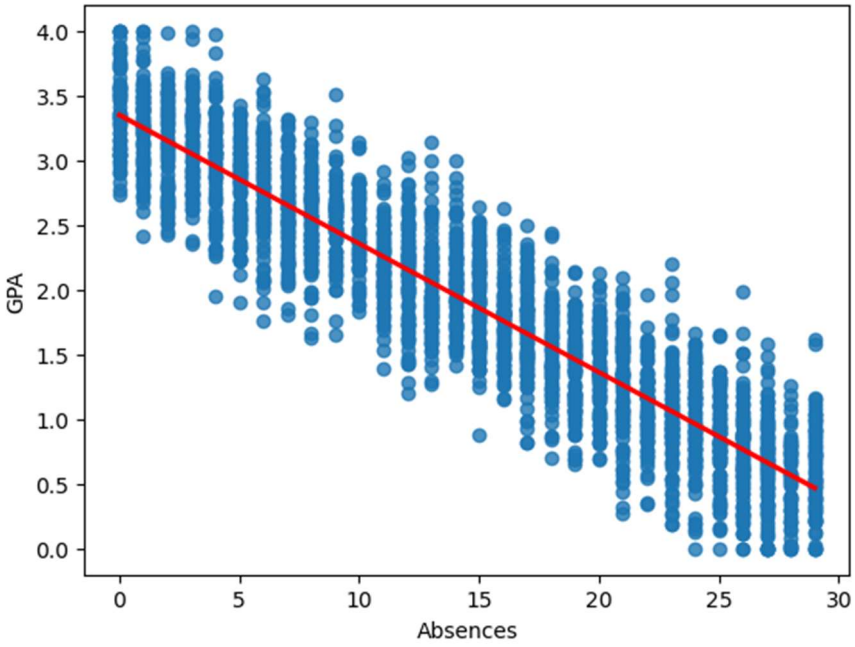


Figure 6 Relation between Absences and GPA

The relationship between Absences and GPA is clearly negative and very linear (Figure 6).

I select the five most important features and build an optimized ETR-model. This model yields a cross-validation score of 0.91 and a 0.91 accuracy on the test. Both scores are a bit worse from the earlier models. I'll plot the real and predicted GPA on all models this far to demonstrate their accuracy:

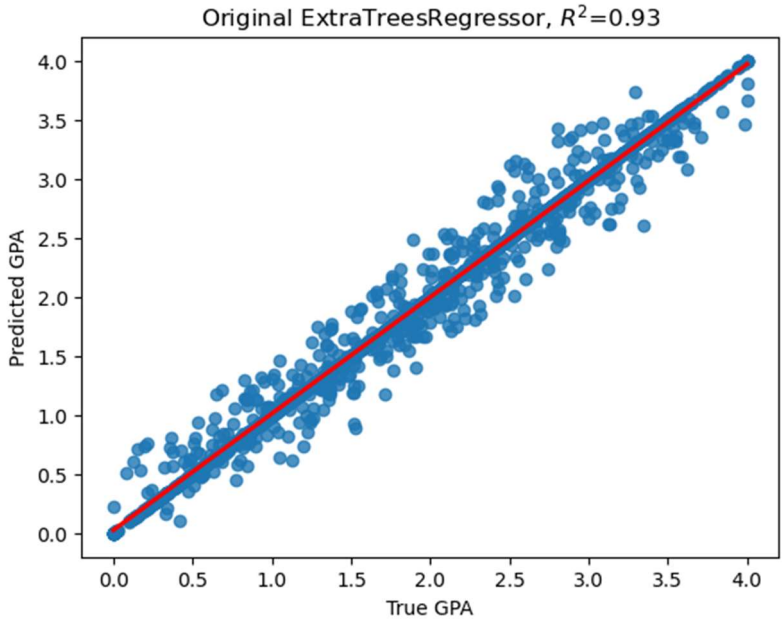


Figure 7 Original ETR true vs. predicted GPA

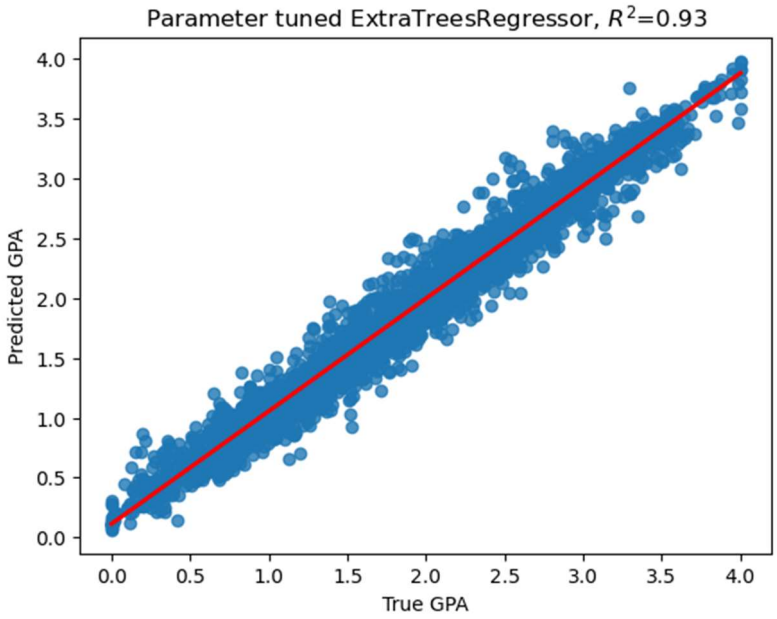


Figure 8 Parameter tuned ETR true vs. predicted GPA

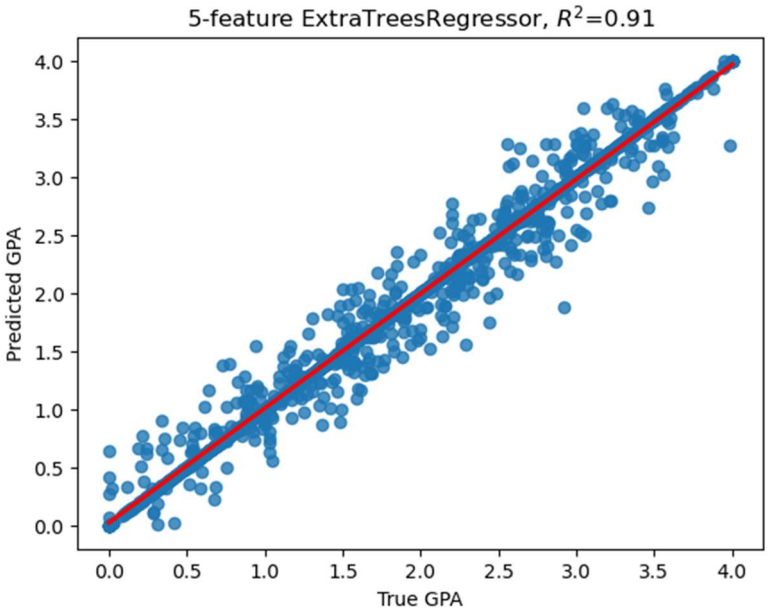


Figure 9 5-feature ETR true vs. predicted GPA

The original ETR (Figure 7) seems to be about 0.02 points more accurate than the optimized one (Figure 9) and about the same as the tuned parameter (Figure 8) model. This time, the full ETR-model and tuned parameters take the win.

To have some fun and to compare the earlier models with Neural Networks, I'll create two ANN models, which regress the GPA with all the features and with the five most important ones. This is also something that Irshada and Kumar (2023) succeeded best with in their research. The ANN yielded an even better accuracy than the Random Forest-model. I'll be using the MLPRegressor-algorithm with hidden layer sizes of 10 and 6 as well as max iterations of 500.

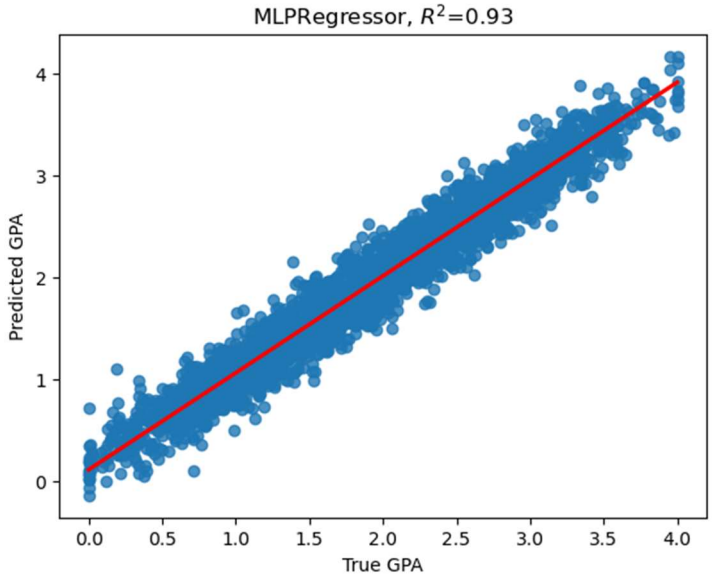


Figure 10 Full-feature ANN true vs. predicted GPA

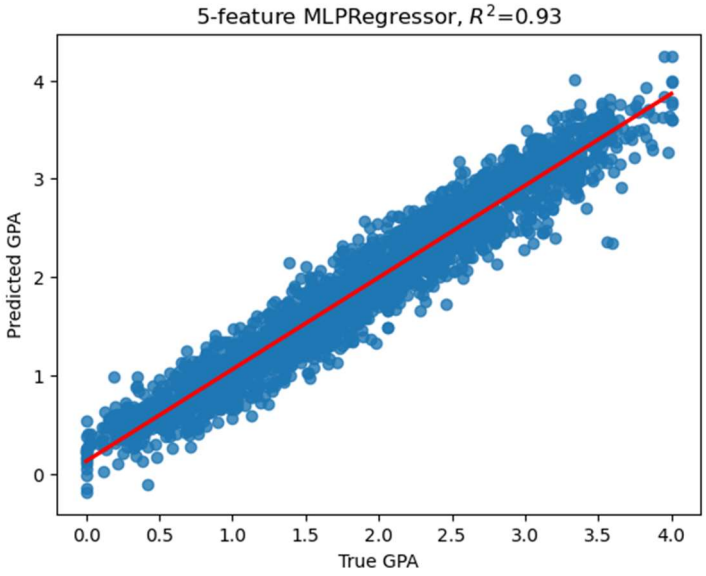


Figure 11 5-feature ANN true vs. predicted GPA

The full feature ANN (Figure 10) yielded a cross-validation score of 0.94 and a and a 0.95 accuracy on the test. The five feature ANN (Figure 11) yielded a cross-validation score of 0.93 and 0.94 accuracy on the test. Thus far, the ANN with all the features has had the best cross-validation and test scores of every model. The full feature ANN seems to be the best predictor as well, although the 5-feature ANN scored very well too.

To further emphasize the linearity of the data, I imported statsmodes.api library and fitted an ordinary linear regression on the data. Here is the summary of the linear regression:

OLS Regression Results

Dep. Variable:	GPA	R-squared:	0.954
Model:	OLS	Adj. R-squared:	0.954
Method:	Least Squares	F-statistic:	4118.
Date:	Tue, 19 Nov 2024	Prob (F-statistic):	0.00
Time:	15:07:37	Log-Likelihood:	502.85
No. Observations:	2392	AIC:	-979.7
Df Residuals:	2379	BIC:	-904.6
Df Model:	12		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	2.5974	0.061	42.407	0.000	2.477	2.717
Age	-0.0055	0.004	-1.542	0.123	-0.013	0.002
Gender	0.0149	0.008	1.854	0.064	-0.001	0.031
Ethnicity	0.0026	0.004	0.669	0.503	-0.005	0.010
ParentalEducation	0.0017	0.004	0.432	0.666	-0.006	0.010
StudyTimeWeekly	0.0289	0.001	40.571	0.000	0.028	0.030
Absences	-0.0998	0.000	-209.572	0.000	-0.101	-0.099
Tutoring	0.2501	0.009	28.473	0.000	0.233	0.267
ParentalSupport	0.1514	0.004	42.188	0.000	0.144	0.158
Extracurricular	0.1907	0.008	23.042	0.000	0.174	0.207
Sports	0.1940	0.009	22.138	0.000	0.177	0.211
Music	0.1431	0.010	14.121	0.000	0.123	0.163
Volunteering	-0.0085	0.011	-0.771	0.441	-0.030	0.013

Omnibus:	5.625	Durbin-Watson:	1.984
Prob(Omnibus):	0.060	Jarque-Bera (JB):	5.682
Skew:	-0.112	Prob(JB):	0.0584
Kurtosis:	2.918	Cond. No.	381.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 12 Linear regression summary

As visible from the summary (Figure 12), the most important variables, especially Absences, can be found as statistically most significant in the linear regression. The R^2 score is also very good, almost 96%. The predictive variables, therefore, capture a very significant amount of the GPA's variance.

3 Conclusion

To conclude the model building, a linear model worked very well this time to predict GPA. Absences was the most important predictor in the data, but it is possible that in other high-school performance data sets it would not achieve such significance. ANNs had the best scores, but they also require more computing power, i. e. give or take. To also summarize the final scores, here is a table of every model:

Model	**Training score**	**CV-score**	**Test score**
Original ETR	**1**	0.9312	0.9252
Tuned params. ETR	0.9719	0.9286	0.9217
5-feature ETR	**1**	0.9090	0.9140
Full feature ANN	0.9498	**0.9411**	**0.9485**
5-feature ANN	0.9360	0.9344	0.9355

Table 1 Model scores

References

Bilal, M., Omar, M., Anwar, W. *et al.* (2022). The role of demographic and academic features in a student performance prediction. *Sci Rep* **12**, 12508. https://doi.org/10.1038/s41598-022-15880-6

Irshada, M., Kumar, V. (2023). SMOTE and ExtraTreesRegressor based random forest technique for predicting Australian rainfall. *Int. j. inf. tecnol.* **15**, 1679–1687https://doi.org/10.1007/s41870-023-01185-y

Kallunki, E. (2023, 16th of March). Koululaisten matematiikan osaaminen on heikentynyt – osalla oppilaista taustalta löytyy puutteellinen sanavarasto. Yle. Retrieved 2024-11-15 from https://yle.fi/a/74-20022436

Oku, A. Y. A., & Sato, J. R. (2021). Predicting Student Performance Using Machine Learning in fNIRS Data. *Frontiers in human neuroscience*, *15*, 622224. https://doi.org/10.3389/fnhum.2021.622224

Opetus- ja kulttuuriministeriö. (2022). PISA-tutkimus ja tulokset 2022. Retrieved 2024-11-15 from https://okm.fi/pisa-2022

Rabie El Kharoua. (2024). Students Performance Dataset [Data set]. Kaggle. https://doi.org/10.34740/KAGGLE/DS/5195702

Salem, R. O., Al-Mously, N., Nabil, N. M., Al-Zalabani, A. H., Al-Dhawi, A. F., & Al-Hamdan, N. (2013). Academic and socio-demographic factors influencing students' performance in a new Saudi medical school. *Medical teacher*, *35 Suppl 1*, S83–S89. https://doi.org/10.3109/0142159X.2013.765551

Välisuo, P. (2023) AML. University of Vaasa. Retrieved 2024-11-16 from https://pevalisuo.github.io/AML/DecisionTrees.html

Attachments

The attachments are a Jupyter Notebook and a csv-file of the original data. They can be found from inside the zip-file.